

Research highlights

- Introduces a bi-level reinforcement learning (RL) framework for rolling bus timetable optimization and station holding control.
- Maps upper-level planned-headway adjustments to rolling planned dispatch headways and launch times.
- Couples asynchronous dispatch planning and holding via a dynamic timetable, holding feedback, and upper-only hindsight credit.
- Reports delay-aware service metrics, including holding time, onboard time, total passenger time, and dispatch lateness.
- Achieves lower mean composite cost than target-headway-only, fixed-timetable, classical holding, and same-simulator RL proxy baselines in the reported ten-seed comparison.